The finiteness half of the mass gap is the nonvanishing of one smeared susceptibility

The requirement $m < \infty$ of the Jaffe–Witten statement reduces to an exact inequality between two ground-state quantities. For any gauge-invariant local scalar φ built from the spatially smeared field, with $\langle\varphi\rangle = 0$,

$$m \;\le\; \frac{g^2\hbar c}{2}\,\frac{\langle\,|D_\varphi|^2\,\rangle_0}{\chi_\varphi}, \qquad D_\varphi = \int d^3y\,\frac{\delta\varphi(y)}{\delta A_i^a(0)}, \qquad \chi_\varphi = \int d^3z\,\langle\varphi(z)\varphi(0)\rangle_0^{\rm c},$$

at fixed regularization, with $g = g(a)$. The continuum version requires a limit of the full weighted numerator and the susceptibility as below. Both sides are intensive: the volume cancels between the numerator and the denominator, which is what makes the zero-momentum operator the right trial excitation and the plaquette of the upper-bound note the wrong one. The inequality is the energy-weighted sum rule, m bounded by the mean excitation energy carried by φ, so its content is: **if some local gauge-invariant observable has nonvanishing zero-momentum fluctuations in the vacuum, the theory has an excitation of finite energy.** Conversely $m = \infty$ forces every such susceptibility to vanish, that is, the vacuum is an eigenstate of every smeared gauge-invariant operator. Spatial gradient flow supplies a candidate smearing; its needed Hamiltonian equal-time continuum estimates remain hypotheses; what is assumed, and stated as hypothesis (F1), is that the equal-time flowed correlators have continuum limits, which is not Lüscher's four-dimensional theorem but its Hamiltonian counterpart. In free Maxwell theory the explicit ratio carries g^{-2}, cancelling the prefactor in the bound, which is $4.26\hbar c/\sqrt{8t}$; since the bound cannot fall below m, the flow time at which that scaling must break down is the correlation length $\hbar c/m$. In free Maxwell theory the scaling persists for all t and the bound correctly tends to $m = 0$. The inequality itself is the textbook single-mode (Feynman–Bijl) bound; what is specific here is the zero-momentum intensive form, the flow as the smearing that makes it cutoff-independent, and the identification with the finiteness half of the Millennium statement. Constants explicit; nothing promoted.

1. The continuum bound

Take the temporal-gauge Hamiltonian in the convention of G07 Proposition 7, with g dimensionless and $[A] = L^{-1}$:

$$H = \int d^3x\,\Big[\frac{g^2c}{2\hbar}\,\Pi_i^a\Pi_i^a + \frac{\hbar c}{4g^2}\,F_{ij}^aF_{ij}^a\Big], \qquad [A_i^a(x),\Pi_j^b(y)] = i\hbar\,\delta^{ab}\delta_{ij}\delta^3(x-y),$$

obtained from $S = -\frac{\hbar}{4g^2}\int F^a_{\mu\nu}F^{a\mu\nu}d^3x\,c\,dt$ with $\Pi_i^a = (\hbar/g^2c)\dot A_i^a$.

Proposition 1. Let O be a real functional of the spatial field A alone, gauge invariant, with $\langle O\rangle_0 = 0$ and $O\Omega \in D(H)$. Then

$$E_1 - E_0 \;\le\; \frac{g^2\hbar c}{2}\,\frac{\displaystyle\int d^3x\,\Big\langle\Big|\frac{\delta O}{\delta A_i^a(x)}\Big|^2\Big\rangle_0}{\langle O^2\rangle_0}.$$

Proof. $O\Omega \perp \Omega$, so by min-max $E_1 - E_0 \le \langle O\Omega, (H-E_0)O\Omega\rangle/\|O\Omega\|^2$, and $2\langle\Omega, O(H-E_0)O\Omega\rangle = \langle\Omega,[O,[H,O]]\Omega\rangle$ since $[O,[H,O]] = 2OHO - O^2H - HO^2$. The magnetic term commutes with O. For the electric term, $[\Pi_i^a(x),O] = -i\hbar\,\delta O/\delta A_i^a(x)$ and

$$[O,[\Pi_i^a\Pi_i^a,O]] = -2\big[\Pi_i^a,O\big]^2 = 2\hbar^2\Big|\frac{\delta O}{\delta A_i^a}\Big|^2,$$

so $[O,[H,O]] = \frac{g^2c}{2\hbar}\cdot 2\hbar^2\int d^3x|\delta O/\delta A|^2 = g^2\hbar c\int d^3x|\delta O/\delta A|^2$. Divide by $2\langle O^2\rangle$. □

Only the electric term contributes: the bound measures how much the trial observable is disturbed by the conjugate field, which is the same quantity that Section 2 of the action-floor note identifies as the carrier of $\hbar$.

2. Zero momentum makes the bound intensive

Let $\varphi(x)$ be a gauge-invariant local scalar with $\langle\varphi\rangle_0 = 0$ and take $O = \int_V d^3y\,\varphi(y)$ in a periodic box of volume V. Write

$$D_i^a(x) = \frac{\delta O}{\delta A_i^a(x)} = \int_V d^3y\,\frac{\delta\varphi(y)}{\delta A_i^a(x)}.$$

By translation invariance of the ground state, $\langle|D(x)|^2\rangle_0$ does not depend on x, so the numerator of Proposition 1 is $V\langle|D|^2\rangle_0$, while $\langle O^2\rangle_0 = \int_V\int_V\langle\varphi(y)\varphi(y')\rangle_0^{\mathrm{c}} = V\chi_\varphi$ with $\chi_\varphi = \int d^3z\,\langle\varphi(z)\varphi(0)\rangle_0^{\mathrm{c}}$. The volume cancels:

Corollary 2. For every such φ and every V,

$$\Delta_V \;\le\; \frac{g^2\hbar c}{2}\,\frac{\langle|D_\varphi|^2\rangle_0}{\chi_\varphi}\,,$$

and if $\Delta_V \to m$ and both ground-state quantities converge, the same bound holds for m.

Dimensions: $\delta/\delta A(x)$ carries $L^{-3}/[A] = L^{-2}$, so $[\langle|D|^2\rangle] = [\varphi]^2L^2$ and $[\chi] = [\varphi]^2L^3$; the ratio is an inverse length and $g^2\hbar c$ times it is an energy. The plaquette of the upper-bound note fails precisely because it is a single cell rather than a zero-momentum sum: its variance is an ultraviolet quantity that vanishes in the continuum limit while its gradient does not.

3. Spectral reading and two checks

Inserting a complete set of eigenstates,

$$\langle\Omega,[O,[H,O]]\Omega\rangle = 2\sum_n(E_n-E_0)\,\big|\langle n|O|\Omega\rangle\big|^2, \qquad \langle O^2\rangle_0 = \sum_n\big|\langle n|O|\Omega\rangle\big|^2,$$

so Proposition 1 states that m is at most the mean excitation energy weighted by the spectral weight of O: the f-sum rule. Two consequences.

Conserved observables. If $[H,O] = 0$ the numerator vanishes and the bound reads $\Delta \le 0$. With a unique ground state this is consistent only if $\langle O^2\rangle_0 = 0$: a conserved gauge-invariant observable cannot fluctuate in a nondegenerate vacuum, since $O\Omega$ would be a second state of energy E_0. The bound therefore detects vacuum degeneracy, and is vacuous exactly in that case.

Free Maxwell. Here $m = 0$ and the bound is a positive number for every flow time; Section 5 shows it tends to zero as the smearing grows, which is the correct behaviour for a gapless theory and a check that the bound is not empty.

4. What must be smeared, and the hypothesis

A local operator at a point has divergent gradient in the continuum, so φ must be built from a smeared field. The gauge-covariant smearing is the gradient flow: on the spatial field at fixed time,

$$\partial_s B_i = D_jG_{ji}, \qquad B_i|_{s=0} = A_i, \qquad G_{ij} = \partial_iB_j - \partial_jB_i + [B_i,B_j],$$

the three-dimensional, equal-time counterpart of Lüscher's flow (arXiv:1006.4518v3, equations (1.1)–(1.2), passage level via the local companion), with smearing radius $\sqrt{8t}$ at flow time t and the lattice form given by his equation (1.4). Take $\varphi = \varphi_t = \frac{1}{4}G_{ij}^aG_{ij}^a\big|_{s=t} - \langle\cdot\rangle_0$.

Lüscher's renormalization statement is for the four-dimensional flow in the Euclidean functional integral: expectation values of local gauge-invariant expressions in the flowed field are finite without further renormalization. The object needed here is the equal-time Hamiltonian correlator of the spatially flowed field, which his theorem does not cover. State it as a hypothesis:

(F1) For fixed flow time $t > 0$, the equal-time ground-state susceptibility $\chi_{\varphi_t,a}$ converges to a finite strictly positive limit χ_t, and the complete weighted numerator $g(a)^2\langle|D_{\varphi_t,a}|^2\rangle_0$ converges to a finite limit K_t, as $a \to 0$ along the scaling curve (after the specified volume limit).

On any fixed lattice both quantities are finite and positive and Corollary 2 holds unconditionally, so (F1) is entirely a statement about the continuum limit. Its first half is the flow's renormalization property in the Hamiltonian framework; its second half, $\chi > 0$, is nontriviality: the vacuum is not an eigenstate of the flowed energy density.

Proposition 3 (the finiteness half). Assume the continuum theory exists with a unique vacuum and translation invariance, that the finite-volume gaps converge to m, and (F1) for one flow time t. Then

$$m \;\le\; \frac{\hbar c}{2}\,\frac{K_t}{\chi_t} \;<\; \infty.$$

This gives the $m < \infty$ clause conditional on the weighted numerator, the positive susceptibility and spectral-limit hypotheses. The remaining clause, $m > 0$, is untouched: nothing above excludes $m = 0$.

5. The free Maxwell check and the coupling correction

The coefficient g^2 in the variational bound is cancelled by the g^{-2} dependence of the ratio of the two free-field observables. Dimensional analysis alone cannot determine this coupling dependence. The following is an exact Gaussian calibration; extending it to an interacting theory near its confinement scale requires nonperturbative control, and an upper bound cannot prove a positive gap.

5a. Free ground state

Take the abelian case of the Hamiltonian of §1,

$$H = \int d^3x \,\Big[\frac{g^2 c}{2\hbar}\,\Pi_i\Pi_i + \frac{\hbar c}{2g^2}\,b_i b_i\Big], \qquad b_i = \varepsilon_{ijk}\partial_j A_k, \qquad [A_i(x),\Pi_j(y)] = i\hbar\delta_{ij}\delta^3(x-y),$$

in Coulomb gauge $\partial_i A_i = 0$, two transverse polarizations. In Fourier modes each transverse mode is an oscillator of mass $M = \hbar/(g^2 c)$ and frequency $\omega_k = ck$, since $\frac{1}{2}M\omega_k^2 = \hbar c k^2/(2g^2)$ matches the magnetic term. Its ground-state width gives

$$\big\langle \hat{A}_i(k)\hat{A}_j(k')\big\rangle_0 = (2\pi)^3\delta^3(k+k')\,P^{\mathrm{T}}_{ij}(k)\,\frac{\hbar}{2M\omega_k} = (2\pi)^3\delta^3(k+k')\,P^{\mathrm{T}}_{ij}(k)\,\frac{g^2}{2k},$$

with $P^{\mathrm{T}}_{ij} = \delta_{ij} - \hat{k}_i\hat{k}_j$. Dimensions: $[\hat{A}] = L^2$, $[(2\pi)^3\delta^3] = L^3$, and $g^2/2k$ is a length.

5b. Equal-time flow

For an abelian field the flow equation of §4 reduces to $\partial_s B_i = \Delta B_i - \partial_i(\partial_j B_j)$, and a transverse initial field stays transverse, so

$$\hat{B}_i(t,k) = e^{-tk^2}\,\hat{A}_i(k), \qquad \hat{b}_i(t,k) = i\varepsilon_{ijl}k_j\,e^{-tk^2}\,\hat{A}_l(k).$$

The flow time has dimension of length squared; the conventional smearing radius is $\sqrt{8t}$. Set $\varphi_t = \frac{1}{2}(b_t\cdot b_t - \langle b_t\cdot b_t\rangle_0)$, of dimension L^{-4}.

5c. Susceptibility

Write $G_{ij}(z) = \langle b_{t,i}(z) b_{t,j}(0)\rangle_0$. Using $\varepsilon_{iab}\varepsilon_{jcd} k_a k_c P^{\mathrm{T}}_{bd}(k) = k^2 P^{\mathrm{T}}_{ij}(k)$ (the longitudinal part of P^{T} drops because $\varepsilon_{iab} k_a \hat{k}_b = 0$),

$$\tilde{G}_{ij}(k) = k^2 P^{\mathrm{T}}_{ij}(k)\, \frac{g^2}{2k}\, e^{-2tk^2} = \frac{g^2 k}{2}\, e^{-2tk^2}\, P^{\mathrm{T}}_{ij}(k).$$

For a Gaussian field Wick's theorem gives $\langle \varphi_t(z)\varphi_t(0)\rangle_0^{\mathrm{c}} = \frac{1}{2}\sum_{ij} G_{ij}(z)^2$ (two pairings), so by Parseval, with $\sum_{ij}[P^{\mathrm{T}}_{ij}]^2 = \operatorname{tr} P^{\mathrm{T}} = 2$,

$$\chi_{\varphi_t} = \int d^3z\, \langle \varphi_t \varphi_t\rangle^{\mathrm{c}} = \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\, \frac{g^4 k^2}{2}\, e^{-4tk^2} = \frac{g^4}{4}\cdot\frac{1}{2\pi^2}\int_0^\infty dk\, k^4 e^{-4tk^2}.$$

With $\int_0^\infty k^4 e^{-\alpha k^2} dk = 3\sqrt{\pi}/(8\alpha^{5/2})$ and $\alpha = 4t$, so $(4t)^{5/2} = 32 t^{5/2}$,

$$\chi_{\varphi_t} = \frac{g^4}{4}\cdot\frac{1}{2\pi^2}\cdot\frac{3\sqrt{\pi}}{256\, t^{5/2}} = \frac{3 g^4}{2048\, \pi^{3/2}\, t^{5/2}}.$$

Dimension: $[\chi] = [\varphi]^2 L^3 = L^{-5}$ and $t^{5/2} \sim L^5$, as required.

5d. Gradient term

By Parseval $O = \int d^3y\, \varphi_t = \frac{1}{2}\int \frac{d^3k}{(2\pi)^3} k^2 e^{-2tk^2} |\hat{A}(k)|^2 + \text{const}$, so

$$D_i(x) = \frac{\delta O}{\delta A_i(x)} = \big(-\Delta\, e^{2t\Delta} A\big)_i(x) = (\nabla \times b_{2t})_i(x), \qquad \hat{D}_i(k) = k^2 e^{-2tk^2}\hat{A}_i(k),$$

the curl of the magnetic field at twice the flow time. Hence

$$\big\langle |D|^2 \big\rangle_0 = \int \frac{d^3k}{(2\pi)^3} k^4 e^{-4tk^2}\cdot 2\cdot\frac{g^2}{2k} = g^2\cdot\frac{1}{2\pi^2}\int_0^\infty dk\, k^5 e^{-4tk^2} = \frac{g^2}{2\pi^2}\cdot\frac{1}{64\, t^3} = \frac{g^2}{128\,\pi^2\, t^3},$$

using $\int_0^\infty k^5 e^{-\alpha k^2} dk = 1/\alpha^3$. Dimension: $[\varphi]^2 L^2 = L^{-6}$ and $t^3 \sim L^6$, as required.

5e. The bound and coupling cancellation

$$\frac{\langle |D|^2\rangle}{\chi} = \frac{g^2}{128\pi^2 t^3}\cdot\frac{2048\,\pi^{3/2} t^{5/2}}{3 g^4} = \frac{16}{3\, g^2\sqrt{\pi t}},$$

$$m \;\le\; \frac{g^2\hbar c}{2}\cdot\frac{16}{3 g^2\sqrt{\pi t}} = \frac{8}{3\sqrt{\pi}}\,\frac{\hbar c}{\sqrt{t}} = \frac{16\sqrt{2}}{3\sqrt{\pi}}\,\frac{\hbar c}{\sqrt{8t}} \approx 4.26\,\frac{\hbar c}{\sqrt{8t}}.$$

Correction. The earlier version inferred $m \le C g^2 \hbar c/\sqrt{8t}$ from dimensional analysis, assigning the pure numbers k_1, k_2 no coupling dependence. The computation shows $\chi \propto g^4$ while $\langle |D|^2\rangle \propto g^2$, so the ratio carries g^{-2} and the bound is of order g^0. The corrected statement is that the perturbative bound is $4.26\,\hbar c/\sqrt{8t}\,[1 + O(g^2)]$, with the coupling entering only through the corrections. The t-dependence $1/\sqrt{8t}$ of the earlier note stands.

5f. Two-photon sum-rule check

By the spectral reading of §3, the bound equals the mean excitation energy weighted by the spectral weight of O. In the free theory O is quadratic in $\hat{A}$ and creates two photons of momenta k and $-k$, of total energy $2\hbar c k$, with weight proportional to $\big[k^2 e^{-2tk^2}\big]^2 \big[g^2/2k\big]^2\, k^2\, dk \propto k^4 e^{-4tk^2} dk$ after the phase-space factor. Therefore

$$\frac{\int_0^\infty 2\hbar c k\cdot k^4 e^{-4tk^2} dk}{\int_0^\infty k^4 e^{-4tk^2} dk} = 2\hbar c\,\frac{1/(64 t^3)}{3\sqrt{\pi}/(256\, t^{5/2})} = 2\hbar c\cdot\frac{4}{3\sqrt{\pi t}} = \frac{8}{3\sqrt{\pi}}\,\frac{\hbar c}{\sqrt{t}},$$

the same number. The agreement checks Proposition 1, the intensive form of Corollary 2 and the Gaussian computation against each other, since both evaluate the same spectral sum rule by different arithmetic routes.

In a weakly coupled regime the free value is a perturbative orientation, not a uniform confinement-scale bound. Its decrease at large flow time is consistent with a gapless photon; monotonic convergence of the interacting spectral mean to the lowest energy is not proved.

6. Status

claim	status
Proposition 1, the bound with explicit constants	proved
Corollary 2, intensive zero-momentum form	proved
f-sum rule reading, conserved-observable check	proved
finiteness of the lattice bound at fixed a	proved
(F1), continuum limit of the flowed equal-time correlators	hypothesis; the flow's renormalization in the Hamiltonian framework
$\chi > 0$	hypothesis; nontriviality
Free Maxwell ratio and two-photon mean	exact calibration, §5
$m > 0$	untouched

The single-mode bound and the use of smeared operators as glueball interpolators are standard; the statement being recorded is the reduction of the Millennium finiteness clause to the positivity of one flowed susceptibility, with the constants and hypotheses written out.

7. Consequence for STATE

The upper side of the problem is now in the same shape as the lower side: an exact inequality at fixed cutoff, plus one renormalization hypothesis for the continuum. The two hypotheses are different in kind. (F1) concerns flowed observables, which are expected to be renormalized by Lüscher's four-dimensional result and whose Hamiltonian counterpart is a bounded, self-contained question; the dressed-vacuum problem of the Schur note concerns the state itself. The next steps this suggests, in order: use the explicit free-field calibration in §5; then state the Hamiltonian flow-renormalization question as its own target, since both the upper-side (F1) and the lower-side dressing are instances of it.